

Date: 4 March, 2026

To

Shampa Banik

Department of Computer Science & Engineering

Bangladesh University of Business and Technology(BUBT)

Subject: Permission to Submit Group Project Proposal

Respected Madam,

We respectfully submit our group project proposal titled "FIRS(Firearms Information Repository System)" for your review and approval. We kindly request your permission to proceed with the project work.

Thank you for your guidance and support.

Sincerely,

Group Members:

- | | |
|--------------------------------|-------------|
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Signature of Supervisor

Bangladesh University of Business and Technology(BUBT)

PROJECT PROPOSAL

On

FIRS(Firearms Information Repository System)

Submission Date: 4th, March, 2026

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Abstract

The Firearm Information Repository System (FIRS) is a government-regulated online system intended to centralize and manage firearm data in Bangladesh. It is intended to be used by citizens, dealers, government agencies, and system administrators to manage licenses, firearm inventory, criminal investigations, dealer transactions, and reporting. The system will be created in a 17-week period following a set structure, and a multidisciplinary team will follow the Software Development Life Cycle (SDLC) approach to ensure a secure, reliable, and efficient system. Some of the recommendations include integration with national identity data, optimizing the system to function during peak hours, and creating a mobile version of the system for field officers, all of which are intended to improve public safety, law enforcement, and efficiency.

Problems

Key problems FIRS addresses:

The Firearm Information Repository System (FIRS) is needed to address the inefficiencies of the current process of managing firearms. The current process of manually managing information on firearms is resulting in loss of information and making information retrieval a slow process.

Currently, there is no centralized information system for tracking information on firearm ownership. This makes tracking information on the transfer of firearms and tracing information on firearms used in crimes a complex process.

The process of issuing licenses and permits is taking too long due to the current manual process and the multiple approvals needed before an applicant is issued a license or permit.

Law enforcement agencies currently do not have the ability to trace information on firearms quickly, while the lack of adequate integration with government databases makes verifying identities and checking for fraud a slow process.

All these issues point to the need for an efficient and centralized information system that can automate the process of managing information on firearms while supporting the work of law enforcement agencies and regulatory agencies.

Objectives

- **Specific:** Digitally manage firearm licenses, dealer permits, inventory, and criminal tracing.
- **Measurable:** Support 500 concurrent users, 2-second response times, ≥99% uptime.
- **Achievable:** Team of 5 developers using SDLC ensures phased completion.
- **Relevant:** Compliance with Bangladesh Arms Act, 1878 and Arms Rules, 1924.
- **Time-bound:** Complete full deployment within 17 weeks.

Existing System Analysis

1. Bangladesh: Digital Arms Management System (DAMS)

Profile:

The Digital Arms Management System (DAMS) is operated by the Public Security Division, which is under the Ministry of Home Affairs. The service enables firearm license holders to renew and check their licenses online.[1]

Key Features:

- Renewal and checking of licenses online
- Login for citizens for checking personal licenses
- Brief information for submitting documents

Shortcomings:

- Does not include firearm ownership records; only licenses.
- Poor linkages with firearm dealers and law enforcement agencies.
- Manual systems still in use, which is a drawback.

Recommendation:

The current system, DAMS, is a starting point but insufficient for a centralized firearm tracking system. FIRS should be implemented, which includes firearm licenses, ownership, and law enforcement tracing.

2. USA: National Instant Criminal Background Check System (NICS)

Profile:

The NICS, run by the FBI, provides instant eligibility checks for prospective gun purchasers in the United States.[2]

Key Features:

- Background checks conducted in real time for criminal, mental, and prohibited person records
- Instant approval or denial for licensed dealers
- High reliability for checking the eligibility of gun purchasers

Shortcomings:

- No tracking of firearm ownership or history
- Background checks only; no tracing of firearm transfers
- Reporting from external agencies; data might be incomplete

Recommendation:

The NICS has shown its effectiveness for checking the eligibility of gun purchasers, and FIRS needs to include modules for ownership, transfer, and permit tracking. The NICS might be included for background checks.

3.International:ArmsTracker (law enforcement/military worldwide)

Profile:

ArmsTracker is a software system geared towards managing the inventory of firearms, ammunition, and explosives for law enforcement and military entities worldwide.[3]

Key Features:

- Centralized firearm inventory system, including chain of custody tracking
- Licensing and registration of authorized personnel
- Reporting and compliance features

Shortcomings:

- Primarily geared towards law enforcement/military, not civilians or license tracking
- In-house deployment may restrict real-time access to the system
- No integration with national citizen records or public license applications

Recommendation:

ArmsTracker is a strong candidate for the inventory and law enforcement components of the FIRS system. For the FIRS, a separate module can be created to manage public licenses, dealer permits, and ownership records, aside from the inventory and law enforcement tracking module of ArmsTracker.

Features of proposed project

Firearm Record Management

Description and Priority:

This feature allows authorized users to manage firearm records, including adding, updating, viewing, and deleting information. Priority: High.

- Benefit: 9
- Penalty if missing: 8
- Cost: 7
- Risk: 5

Functional Requirements:

- REQ-1: Allow users to add new firearm details, ensuring all required fields are completed.
- REQ-2: Enable users to update existing firearm information when changes occur.
- REQ-3: Allow authorized users to delete firearm records, with confirmation prompts.
- REQ-4: Validate all inputs and display clear error messages for incorrect entries.
- REQ-5: Maintain a complete audit trail of all changes to firearm records.

User Authentication and Role Management

Description and Priority:

This feature ensures only authorized personnel can access the system and enforces role-based permissions. Priority: High.

- Benefit: 9
- Penalty if missing: 9
- Cost: 6
- Risk: 8

Functional Requirements:

- REQ-6: Users must log in using a username and password.
- REQ-7: Passwords must be encrypted for security.
- REQ-8: System access must be restricted based on user roles (Admin, Special, General).
- REQ-9: Admins can add, update, or remove user accounts.
- REQ-10: Log and monitor unauthorized login attempts.

Search and Reporting

Description and Priority:

This feature enables users to efficiently search firearm records and generate reports. Priority: Medium.

- Benefit: 7
- Penalty if missing: 6
- Cost: 5
- Risk: 5

Functional Requirements:

REQ-11: Allow users to search firearm records using keywords.

REQ-12: Provide filters to narrow search results by category, date, or owner.

REQ-13: Generate printable reports of search results or record summaries.

REQ-14: Display results quickly and accurately with error handling for invalid queries.

Other Nonfunctional Requirements

Performance Requirements

- The system shall respond to user requests within 2 seconds under normal conditions.
- The system shall support up to 500 concurrent users.
- Firearm record searches shall return results quickly using indexed data.
- The system shall maintain 99% availability with minimal downtime.

Safety Requirements

- Access to firearm information shall be limited to authorized users only.
- The system shall prevent unauthorized modification or deletion of records.
- Regular data backups shall be performed to prevent data loss.
- All system activities shall be logged for monitoring and accountability.
- The system shall comply with applicable firearm data handling regulations.

Security Requirements

- Users shall be authenticated using secure login credentials.

- Role-based access control shall restrict system functions based on user roles.
- Sensitive data shall be encrypted during storage and transmission.
- The system shall maintain audit logs of user access and activities.
- Security incidents shall be detected and reported promptly.

Software Quality Attributes:

- Reliability: System uptime $\geq 99\%$.
- Maintainability: Code organized for easy updates.
- Portability: Operable on Windows, Linux & Android.
- Testability: Each module can be tested independently.
- Usability: Interfaces must be user-friendly and intuitive.

Development Model

Chosen Development Model: Software Development Life Cycle (SDLC) - Waterfall Model

Based on Waterfall model

Reasons for Choosing this Development Model (Listed Explanation)

1. Clear Requirements

The Firearm Information Repository System (FIRS) is based on established firearm regulations and firearm licensing laws. Since formal requirements are established in law, a structured sequential model is appropriate.

2. Strong Documentation Requirement

Government systems require complete documentation for auditing, compliance verification, and legal traceability. Waterfall Model enforces documentation at every phase.

3. Formal Approval at Every Phase

Every phase (Requirements, Design, Implementation, Testing, Deployment) requires approval before moving to the next phase. This is in line with how governments control their projects.

4. High Security and Compliance Sensitivity

The system handles firearm information that is highly sensitive in nature. A structured SDLC model ensures that security design, validation, and verification are performed systematically.

5. Clear Phase Separation

The model separates the phases in a distinct manner:

- Requirement Analysis
- System Design
- Implementation
- Testing
- Deployment
- Maintenance

6. Reduced Scope Creep

The system minimizes the possibility of uncontrolled changes that may compromise the integrity of firearm laws and regulations.

7. Audit and Accountability Support

The system supports accountability through formal phase-wise deliverables (SRS, Design Documents, Test Reports).

Solution Overview

The Firearm Information Repository System (FIRS) is designed to centralize firearm licensing, dealer regulation, weapon transaction tracking, and compliance monitoring under a secure digital framework.

The system ensures controlled access for customers, dealers, authorities, law enforcement, and administrators through structured workflows and role-based permissions.

Two implementation approaches are proposed: a government-hosted centralized system and a scalable cloud-based national registry.

Solution 1: Centralized Web-Based Government Controlled System

A centralized web application hosted on the government servers with the ability to manage firearm licenses, dealer permits, weapon transactions, compliance checks, and tracking of criminals.

How It Works

- Customers apply for a license through the web application.

- Dealers report the sale of firearms through the web application.
- Government authorities approve or deny the license applications.
- Authorized personnel search for the records of the weapons.
- Administrator maintains the data.

Major Building Blocks

- Web Application (Front End Interface)
- Application Server (Business Logic Layer)
- Centralized Database Server
- Role-Based Access Control (RBAC)
- Reporting & Monitoring Module
- Security Layer

Solution 2: Cloud-Based Secure National Firearm Registry

This solution involves a cloud-based system with government-approved infrastructure for national-level scalability with distributed access.

How This Solution Works

- All stakeholders can access this system through a cloud-based portal.
- Synchronizes sales and transfers in real-time.
- Automates license eligibility validation.
- Provides a data analytics tool for authorities

Major Building Blocks of This Solution

- Cloud Infrastructure
- REST API Layer
- Encrypted Database
- Dealer Integration Module
- Compliance & Audit Engine
- Backup & Disaster Recovery System

a)DFD

Process P0: Firearm Information Repository System (FIRS)

Central system for firearm registration, licensing, transactions, compliance, and reporting.

External Entities :

1. Customer

- License Application
- License status & Weapon Info.

2. Authorized Dealer

- Sales & Transfer Data
- Weapon Status

3. Govt. Authority

- License Approval
- Monitoring

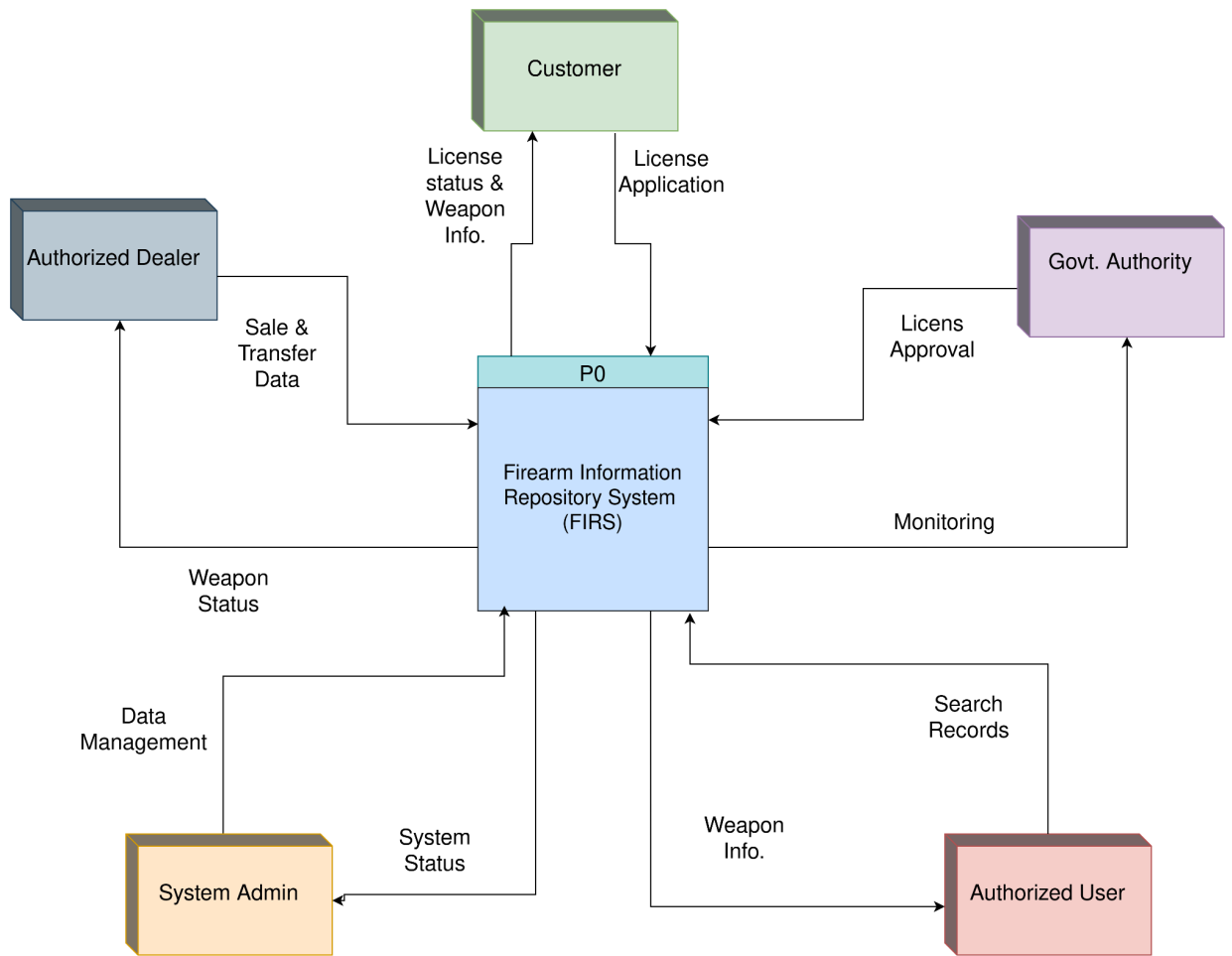
4. Authorized User

- Search Records
- Weapon Info.

5. System Admin

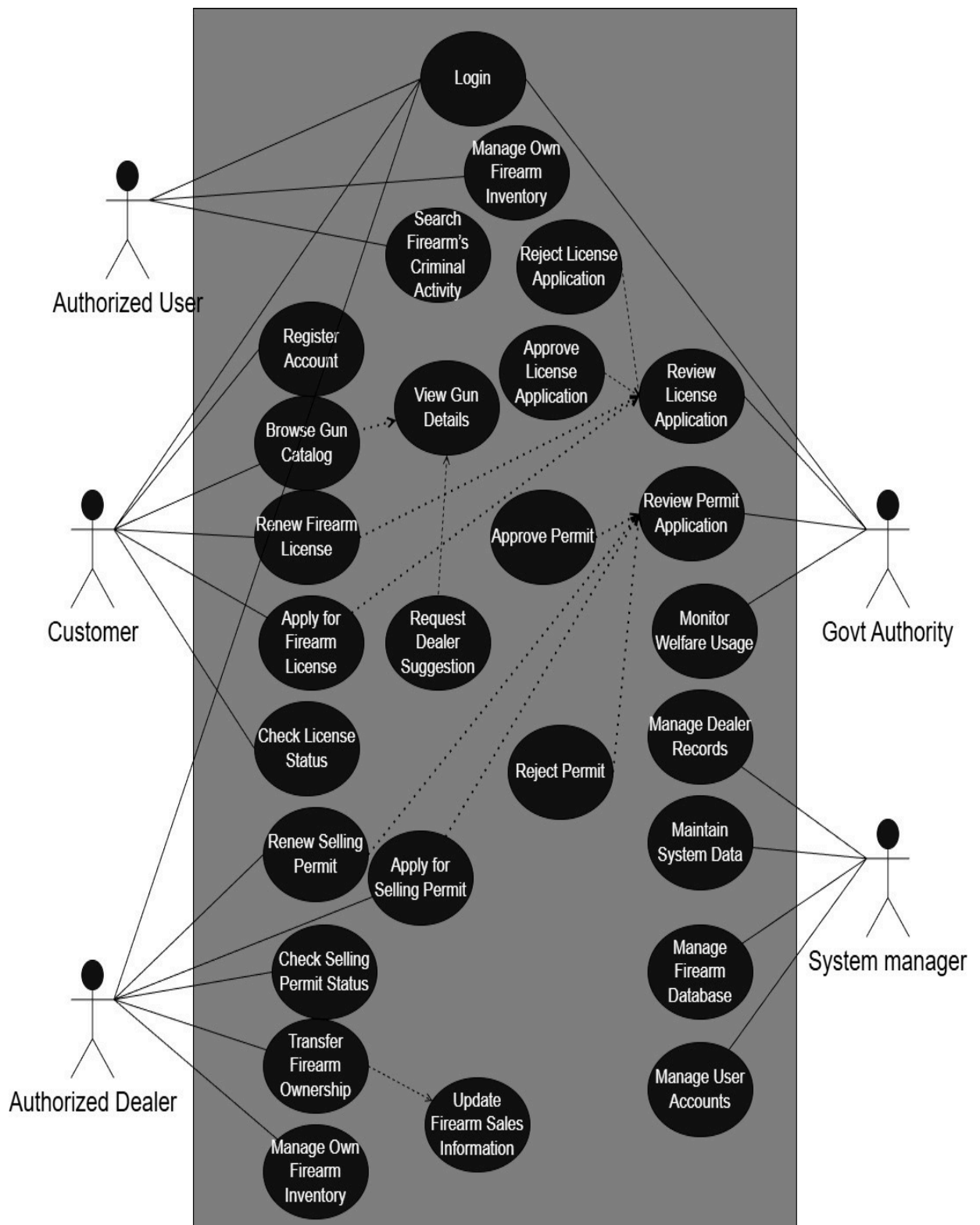
- Data Management
- System Status

The Context level DFD of FIRS



Entities, Types & Relationships

b) Use Case Diagram



Use Case Descriptions

1) Register / Login

Basic:

Choose Register/Login → Enter email & password → System verifies credentials → User authenticated.

Alternate:

- Wrong password → Show error + reset option
- Duplicate email → Suggest Login

2) Apply for Firearm License

Basic:

Login → Open license application → Fill personal & firearm details
→ Upload documents → Submit application.

Alternate:

- Missing documents → Prompt user to upload
- Invalid data → Show validation error

3) Renew Firearm License

Basic:

Login → Select renew license → Verify existing license → Submit
renewal request.

Alternate:

- License expired beyond renewal period → Show rejection message

4) Check License Status

Basic:

Login → Open license status → System retrieves application status
→ Display result.

Alternate:

- No application found → Show notification

5) Browse Gun Catalog

Basic:

Open catalog → System displays available firearms → User browses
list.

Alternate:

- No firearms available → Show empty catalog message

6) View Gun Details

Basic:

Select firearm → System shows specifications, price, and
availability.

Alternate:

- Firearm record unavailable → Show error message

7) Request Authorized Dealer Suggestion

Basic:

View gun details → Request dealer suggestion → System lists authorized dealers with stock.

Alternate:

- No dealer available → Show unavailability message

8) Apply for Selling Permit

Basic:

Dealer login → Open permit application → Enter business details → Submit request.

Alternate:

- Incomplete business info → Prompt correction

9) Renew Selling Permit

Basic:

Dealer login → Select permit renewal → Confirm details → Submit renewal.

Alternate:

- Permit expired beyond allowed period → Show renewal denial

10) Check Selling Permit Status

Basic:

Dealer login → View permit status → System displays current status.

Alternate:

- Permit not found → Show alert

11) Review License Application

Basic:

Authority login → Open pending applications → Review documents & details.

Alternate:

- Incomplete application → Mark for resubmission

12) Approve License Application

Basic:

Review application → Approve request → System updates license status.

Alternate:

- Legal issue detected → Redirect to rejection

13) Reject License Application

Basic:

Review application → Reject request → System records reason & notifies user.

Alternate:

- Missing rejection reason → Prompt authority to provide one

14) Review Selling Permit Application

Basic:

Authority login → Open permit requests → Verify dealer compliance.

Alternate:

- Missing documents → Request resubmission

15) Approve Selling Permit

Basic:

Review permit → Approve → System activates dealer permit.

Alternate:

- Compliance issue found → Redirect to rejection

16) Reject Selling Permit

Basic:

Review permit → Reject → System records reason & notifies dealer.

Alternate:

- Rejection reason missing → Prompt authority

17) Update Firearm Sales Information

Basic:

Dealer login → Enter sale details → Submit update → System saves record.

Alternate:

- Invalid firearm ID → Show error

18) Transfer Firearm Ownership

Basic:

Dealer login → Select firearm → Verify buyer license → Confirm transfer.

Alternate:

- Buyer license invalid → Block transfer

19) Search Firearms Related to Criminal Activity

Basic:

Authorized user login → Enter search criteria → System returns matching records.

Alternate:

- No match found → Show empty results

20) Monitor Welfare Usage

Basic:

Authority login → Access usage reports → Review compliance data.

Alternate:

- Data unavailable → Show system notice

21) Manage User Accounts

Basic:

Admin login → View users → Add/Update/Deactivate accounts.

Alternate:

- Invalid role assignment → Show error

22) Manage Firearm Database

Basic:

Admin login → Access firearm records → Add/Edit/Delete data.

Alternate:

- Duplicate firearm entry → Block insertion

23) Manage Dealer Records

Basic:

Admin login → Access dealer data → Update or verify records.

Alternate:

- Dealer record not found → Show error

24) Maintain System Database

Basic:

Admin login → Run maintenance tasks → Backup and optimize database.

Alternate:

Maintenance failure → Log error & notify admin

c)ERD

i)Entities & Attributes:

1. Firearm

- Firearm_ID (PK)
- Serial_Number
- Type
- Model
- Status

2. Supplier

- Supplier_ID (PK)
- Name
- Contact_Number
- Address

3. Authorized_User

- User_ID (PK)
- Name
- Rank
- Department

4. Issue_Record

- Issue_ID (PK)
- Issue_Date
- Return_Date
- Firearm_ID (FK)
- User_ID (FK)

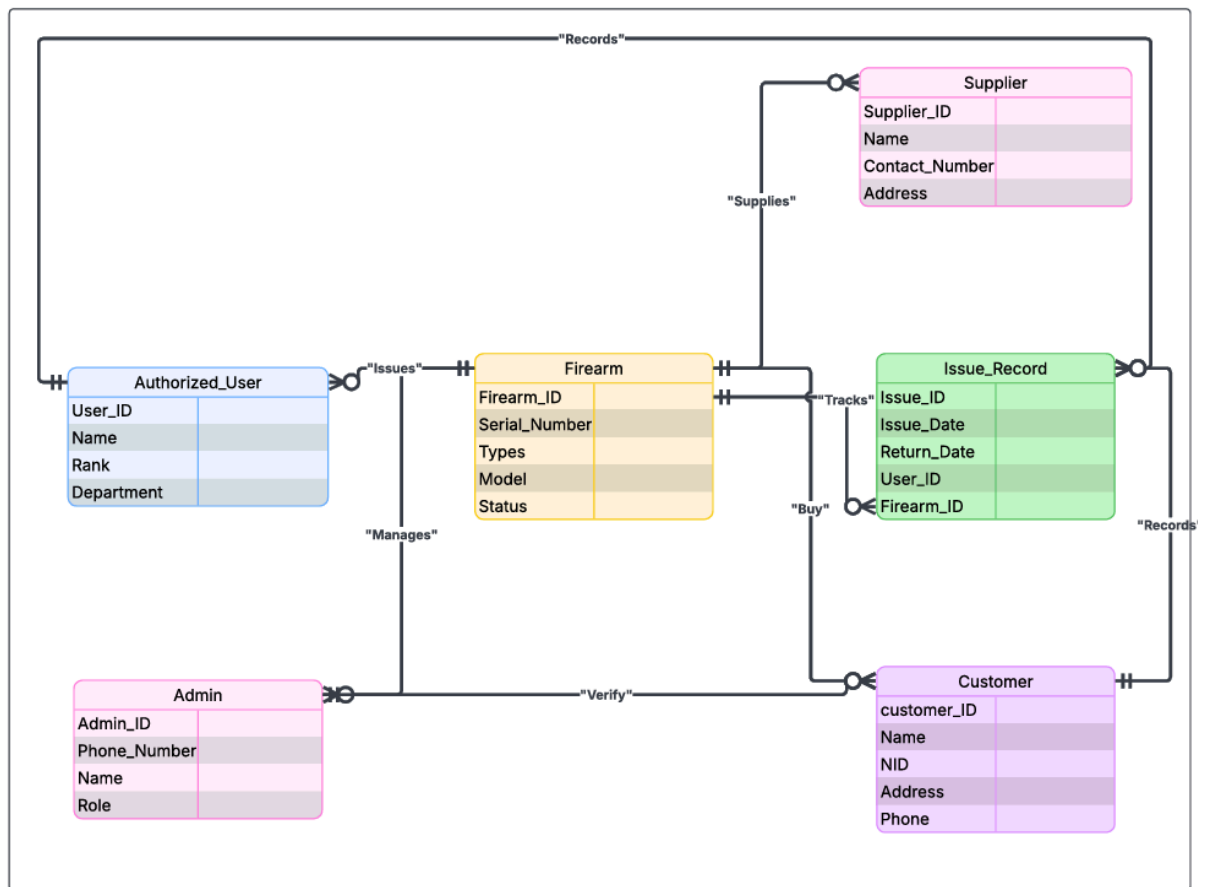
5. Administrator

- Admin_ID (PK)
- Name
- Role

6. Customer

- NID(PK)
- Name
- Addres
- Phone

ii)The ER -Diagram of FIRS



Requirements

1. Hardware Requirements

- Server:
 - CPU: Quad-core or higher, 3.0 GHz+
 - RAM: 16-32 GB
 - Storage: 1-2 TB SSD (RAID recommended for redundancy)
 - Network: 1 Gbps Ethernet, high availability internet connection
- Client Machines (Users, Dealers, Authorities):
 - CPU: Dual-core or higher
 - RAM: 8-16 GB
 - Storage: 256 GB SSD minimum
 - Display: 1080p or higher

- Backup Systems:
 - External storage or cloud backup (encrypted)
 - UPS / power redundancy

2. Software Requirements

- Server OS: Windows Server 2022 / Ubuntu 22.04 LTS / CentOS 9
- Database: MySQL 8 (or equivalent relational DBMS)
- Backend Framework: Java + Spring Boot
- Frontend Framework: React.js, supported by HTML, CSS, JavaScript
- Web Server: Apache, Nginx, or IIS
- Security Tools: SSL/TLS certificates, antivirus/firewall software
- Backup & Monitoring Tools: Automated backup scripts, monitoring dashboard (Nagios, Zabbix, or equivalent)

3. Personnel Requirements

- Project Manager / Team Leader - Oversees planning, schedule, and compliance
- Frontend Developer(s) - Build interfaces using React.js, HTML, CSS, and JS
- Backend Developer(s) - Handle Java + Spring Boot modules, APIs, and business logic
- Database Administrator (DBA) - Design and maintain secure MySQL database
- Quality Assurance / Tester - Test modules for functionality, performance, and security
- Documentation Manager - Prepare SRS, manuals, and audit documentation
- Security Specialist (Optional) - Monitor vulnerabilities, enforce encryption/authentication

4. Tools & Technology

- Version Control: Git + GitHub / GitLab / Bitbucket
- Project Management: Jira, Trello, or MS Project
- Design & Prototyping: Figma, Adobe XD

- Testing Tools: Selenium, Postman, JMeter
- IDE / Editors: VS Code, IntelliJ IDEA (for Java/Spring Boot), PhpStorm (if needed for full stack testing)
- Communication: Slack, Teams, or email

5. Programming Languages

- Frontend: HTML5, CSS3, JavaScript (ES6+), React.js
- Backend: Java + Spring Boot
- Database Queries: SQL (MySQL), optional stored procedures for business logic
- Scripting / Automation: Bash, PowerShell, or Python scripts for maintenance

6. Network & Security Requirements

- Role-based access control (Admin, Authority, Dealer, Citizen)
- Encrypted communication (HTTPS, SSL/TLS)
- Regular backups and disaster recovery plan
- Audit logging for all database and system activity
- Firewall, antivirus, and intrusion detection

7. Other Requirements

- Compliance with Bangladesh Arms Act, 1878 and Arms Rules, 1924
- Scalable architecture for future modules (mobile app, AI analytics, integration with national ID database)
- User-friendly interface for multiple user types
- Support for concurrent users (up to 500 initially)

Budget

PROJECT TITLE

Firearm Information Repository
System (FIRS)

PROJECT MANAGER

Abu Ahmad Mahfuzor Rahman

WBS NUM BER	TASK TITLE	TASK OWNERS	HOURS		AVG RATE	COST	
			MIN	MAX		MIN	MAX
1	Concept developme nt	Abu Ahmed Mahfuzur Rahman, Md Fahim Islam , Md. Mahfuzul Bashar Shaikot	20	30	26	400	900
2	Planning	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	30	34	36	1000	1300
3	Requireme nts analysis	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	20	30	40	800	1200
4	Design	Al Adid, Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Abu Ahmad Mahfuzor Rahman, Md. Sakibur Rahman	50	60	30	1500	1800
5	Developme nt	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	2200	2400	30	66000	72000
5.1	Deliverable #1	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot	1400	1500	35	49000	52500
5.2	Deliverable #2	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	1300	1400	37	48100	51800

5.3	Deliverable #3	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	1600	1700	30	48000	51000
5.4	Deliverable #4	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	1800	1900	37	66600	70300
5.5\	Deliverable #5	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot, Md. Sakibur Rahman	80	100	40	3200	4000
6	Reporting and Measurement	Md Fahim Islam, Md Mahfuzul Bashar Shaikot	200	240	35	7000	8400
7	Implementation	Abu Ahmad Mahfuzor Rahman, Al Adid	50	60	25	1250	1500
8	Quality assurance	Abu Ahmad Mahfuzor Rahman, Al Adid, Md Fahim Islam, Md Mahfuzul Bashar Shaikot, Md Sakibur Rahman	400	500	28	11200	14000
9	QA Testing	Abu Ahmad Mahfuzor Rahman, Al Adid	600	700	33	19800	23100
10	Maintenance	Md Sakibur Rahman, Md Fahim Islam, Md Mahfuzul Bashar Shaikot	140	150	35	4900	5250
11	Launch	Al Adid, Md Sakibur Rahman, Md Fahim Islam, Md Mahfuzul Bashar Shaikot, Abu Ahmad Mahfuzor Rahman	130	135	98	12000	14000

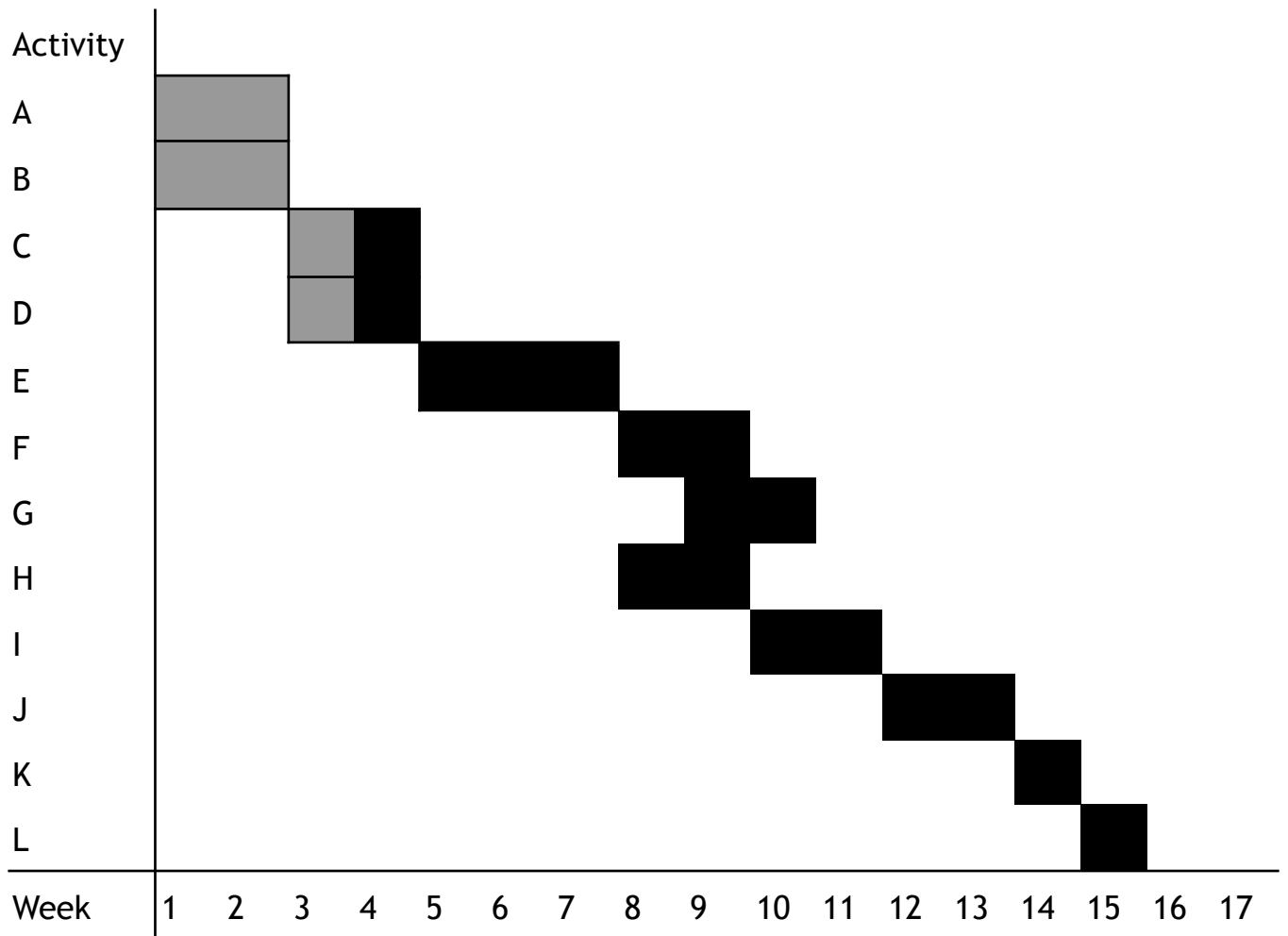
12	Close down	Md Fahim Islam, Md. Mahfuzul Bashar Shaikot,Md. Sakibur Rahman	35	55	61	2450	3000
Totals			1005 5	1099 4	34	34320 0	37605 0

Activity And Time Schedule

a)Project Breakdown Table

Phase	Detailed Task	Predecessor	Duration (Weeks)
Analysis	A Define system objectives	-	2
	B Study of laws & regulations	-	2
	C Requirement analysis & SRS	A	2
	D Feasibility & compliance study	B	2
Development	E System & database design	C	3
	F User authentication module	E	2
	G License management & inventory module	F	2
	H Dealer transaction module	E	2
	I Search & reporting module	H	2
Implementation	J System integration	G	2
	K Testing & validation	J	1
	L Deployment & maintenance	K	1

b) Gantt Chart



Development Team & Stakeholders

ID	Name	Designation	Responsibilities
20245103007	Al Adid	Documentation Manager/ Tester	Google Docs, Overleaf, JUnit
20245103035	Md Fahim Islam	Backend Developer	JAVA, Spring boot
20245103038	Md. Mahfuzul Bashar Shaikot	Frontend Developer	HTML, CSS, JS, React.js
20245103091	Md. Sakibur Rahman	Database Manager	My SQL
20245103092	Abu Ahmad Mahfuzor Rahman	Team Leader / Product Manager	Team managing , Co-coordinating among members and organize the project works to achieve the goals

System Analysts Recommendation

The optimal solution is already provided. Since it is a simple implementation, it is feasible and should be approved for development.

References

- [1]. Public Security Division (PSD), Ministry of Home Affairs, Government of Bangladesh, "Arms License Management System," Available: <http://arms.psd.gov.bd>. [Accessed: Mar. 3, 2026].
- [2]. Federal Bureau of Investigation (FBI), "National Instant Criminal Background Check System (NICS)," Available: <https://www.fbi.gov/services/cjis/nics>. [Accessed: Mar. 3, 2026].
- [3]. ArmsTracker, "ArmsTracker - Firearms, Ammunition and Explosives Inventory Management System," Available: <http://www.armstracker.org>. [Accessed: Mar. 3, 2026].